Deep Learning for Time Series

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Time Series (TS)

Some Examples

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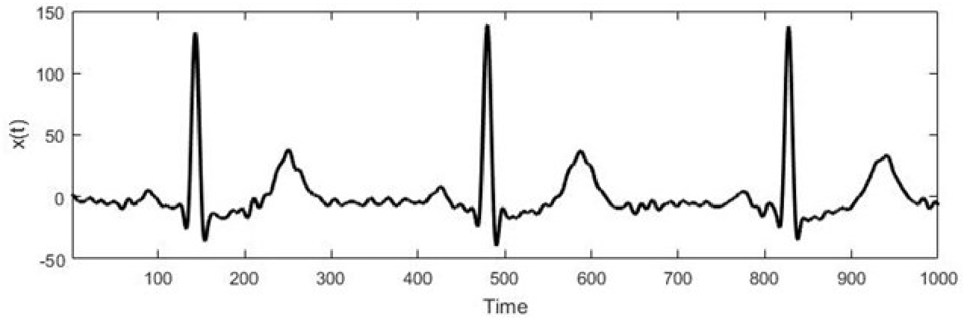
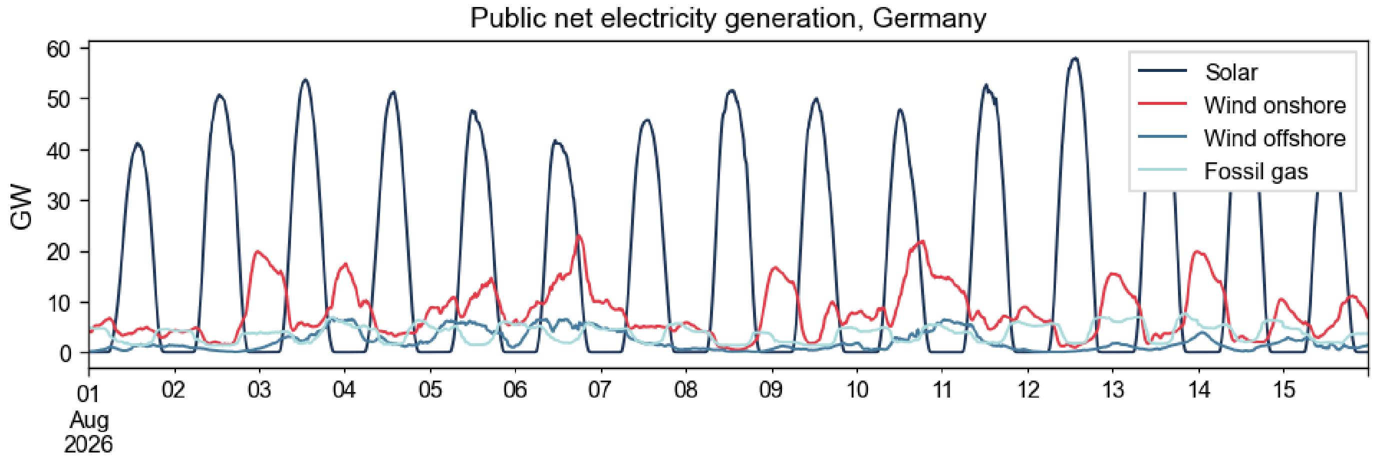


P. Čepulionis and K. Lukoševičiūtė, "Electrocardiogram time series forecasting and optimization using ant colony optimization algorithm," *Mathematical Models in Engineering*, Vol. 2, No. 1, pp. 69–77, Jun. 2016.

Sepp Hochreiter ©NXAI

The language of machines.



What is a Time Series?

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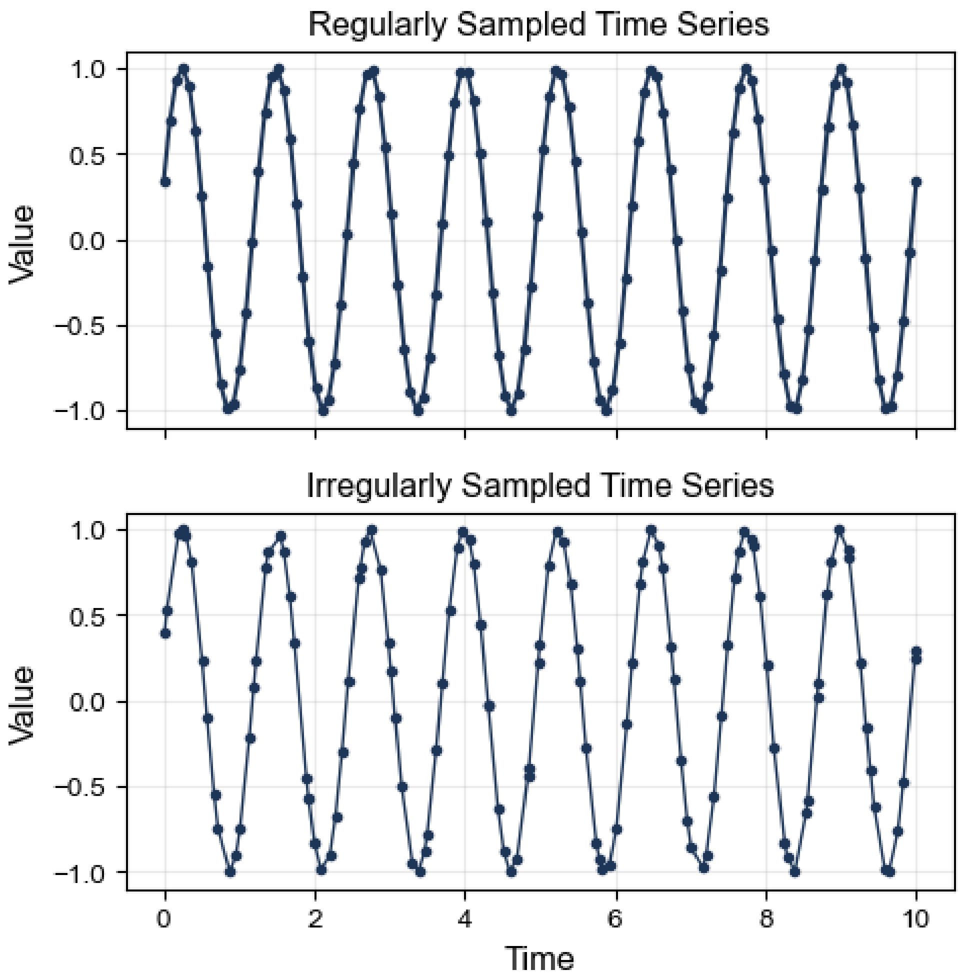
- **Time series**: a sequence of observations recorded at successive points in time
  - Real world: continuous data
  - Digital world: discrete-time data

- **Regular sampling**: constant time increments $\Delta\tau$ between measurements. *Considered hereafter*

- **Irregular sampling**: measurement at non-uniform or unequal time intervals
  - Event-driven measurements: patient visits doctor; stock prices recorded as individual trades happen; …
  - Sensor faults: shut-off; network connection missing; …



Uni- and Multivariate Sequences

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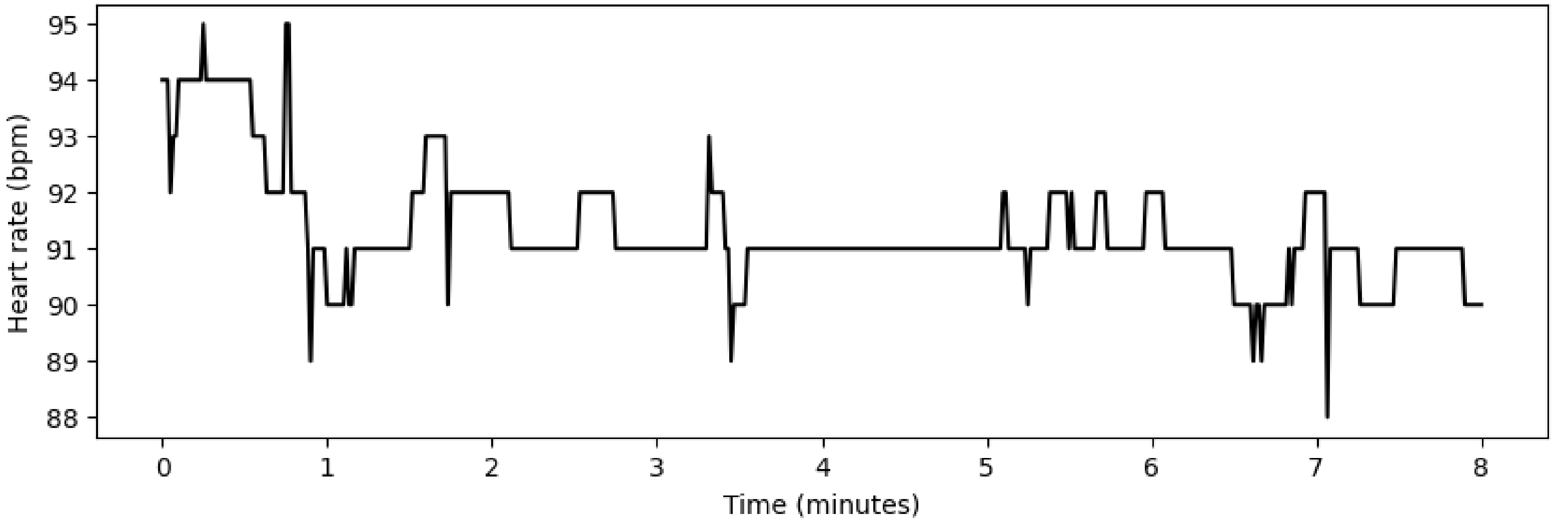
- **Univariate time series**: single-channel measurement,
  e.g. basic fitness tracker records only the patient's heart rate every minute



PhysioNet, bidmc dataset

first patient's data

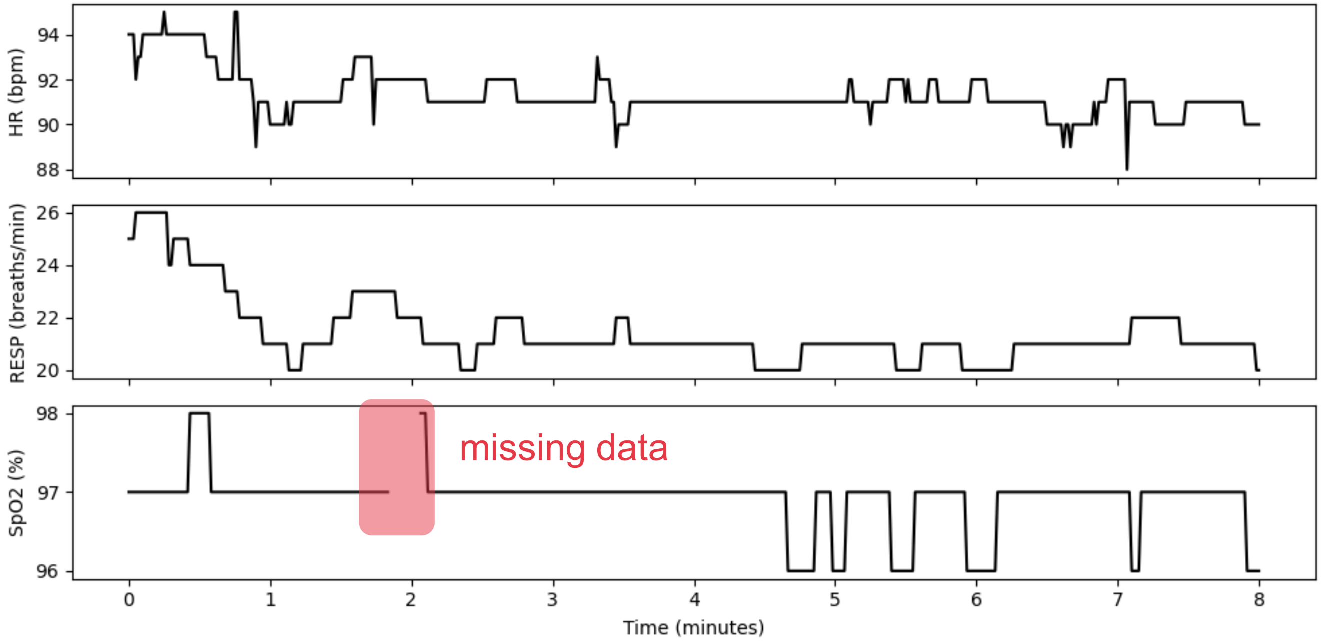
PhysioNet
Maintained by the MIT Laboratory for Computational Physiology

- Mathematical notation: $\boldsymbol{x} = [x_1, x_2, \dots, x_T]^\top \in \mathbb{R}^{T \times 1}$
- Physical time: $\tau_t = \tau_0 + t\Delta\tau$

T: number of time steps
t: index of time step $t = 1, \dots, T$

Uni- and Multivariate Sequences

- **Multivartiate time series**: multi-channel measurement,
  e.g. advanced instrument recording heart rate, respiratory rate and oxygen saturation every minute



PhysioNet, bidmc dataset

first patient's data

PhysioNet
Maintained by the MIT Laboratory for Computational Physiology

T: number of time steps
t: index of time step $t = 1, \dots, T$

C: number of channels
c: index of time steps $c = 1, \dots, C$

- Mathematical notation: $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \dots, \boldsymbol{x}_T]^\top \in \mathbb{R}^{T \times C}$

  $x_{t,c}$ value of channel c at time step t

What makes TS so special?

- **Order matters**: $[x_1, x_2, x_3] \neq [x_2, x_3, x_1]$
  - Shuffling time steps destroys information
  - Order of measurements is not interchangeable!

Latent dimension = time

- The **future must not leak into the past**
  - a model may only use information that would genuinely have been available at that moment
  - Data splitting requires care

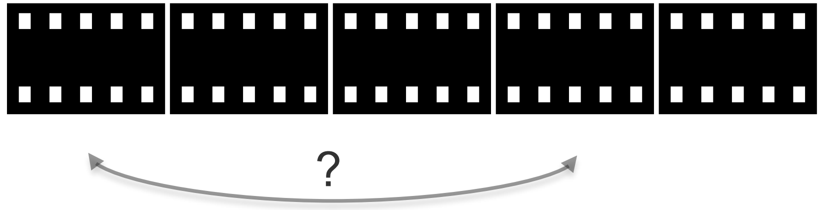
Examples:

[I | live | in | Munich | but | work | in | Berlin]

?

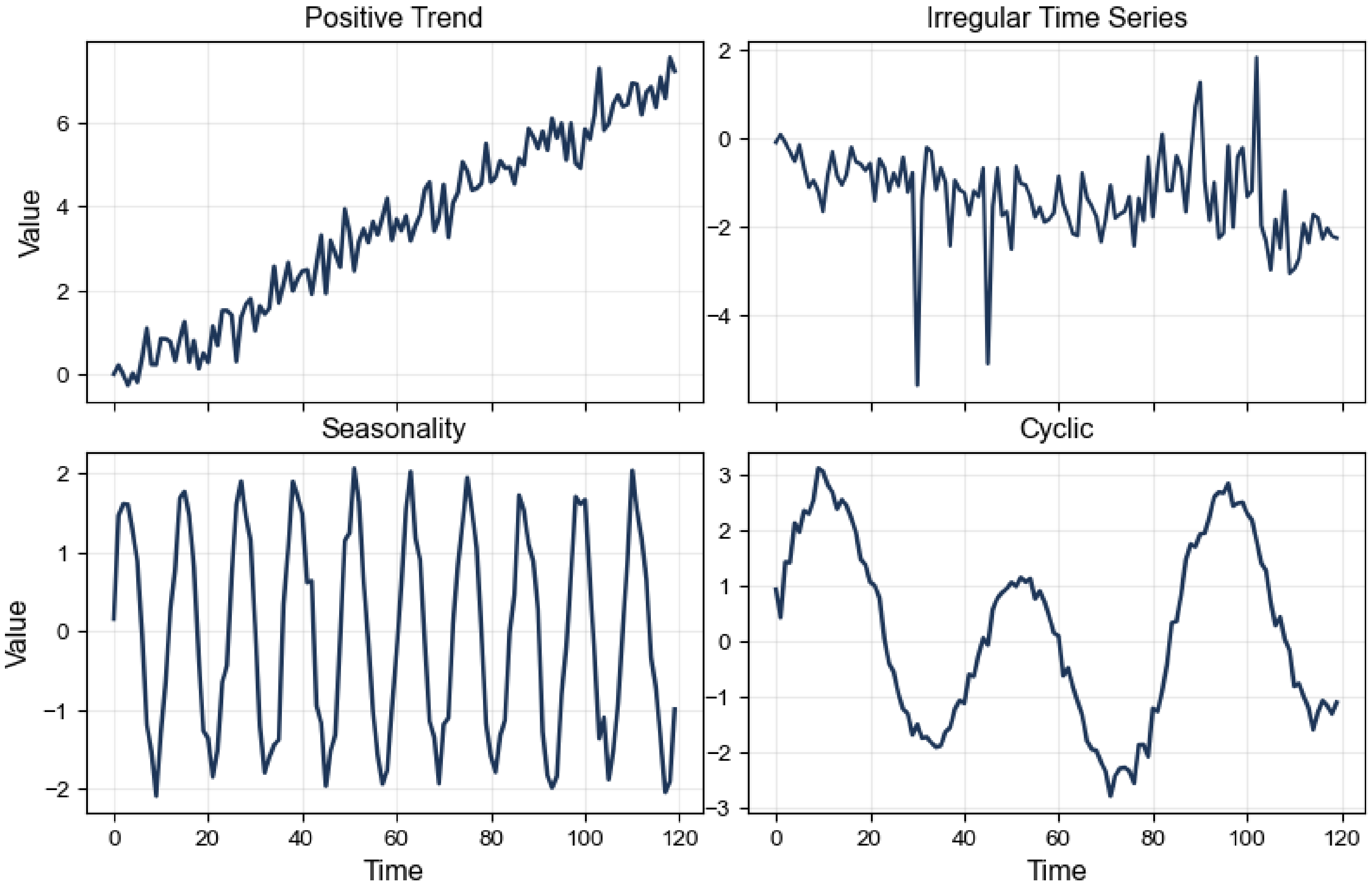
- **Statistical behavior changes over time**
  - Trends, seasonality, transients

- Relevant information exists at **different time scales**
  - Relevant information stored in last sec/min/… year



Characteristics

- Seasonality
- Cyclicality
- Trend
- (Temporal) Irregularity



Seasonality

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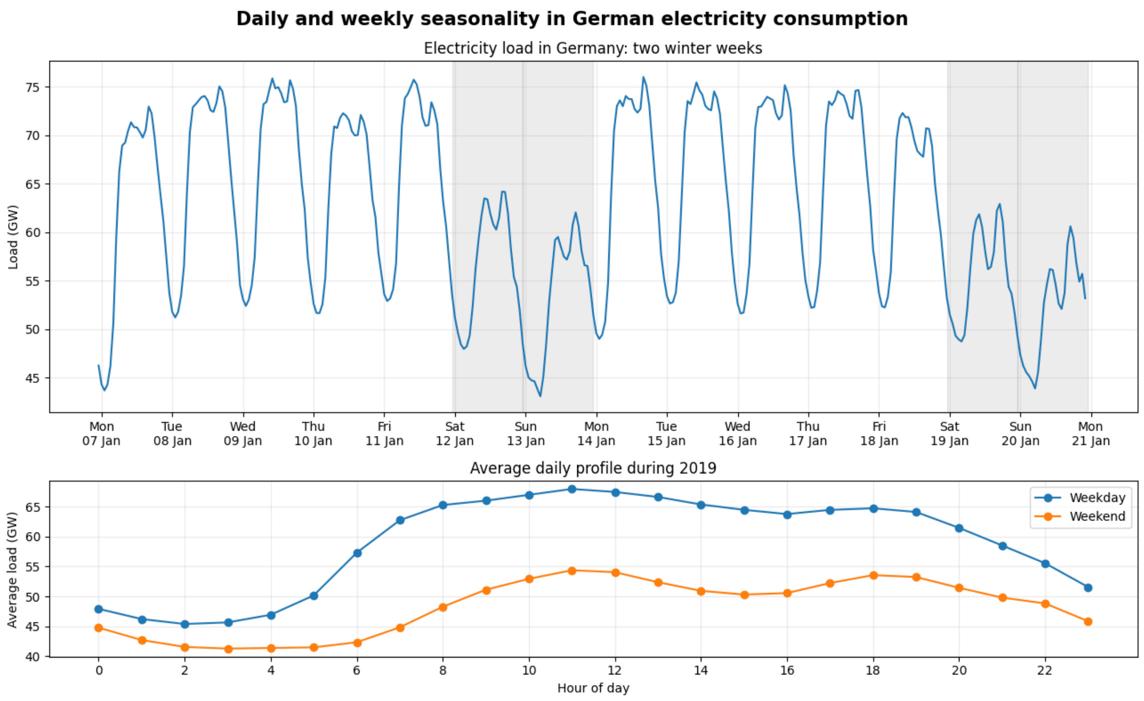
- A **pattern repeats at a known, regular interval** S $\qquad x_t \approx x_{t-S}$
  - not limited to annual cycles
  - several seasonalities can occur simultaneously (single vs. multiple seasonality)

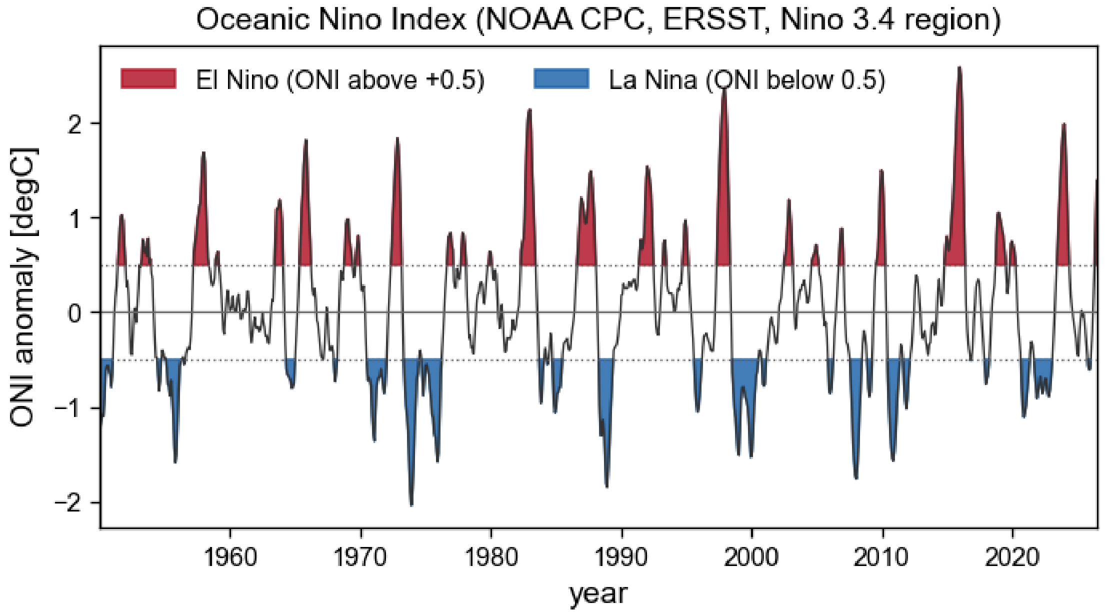
- Examples with seasonal period
  - Household power consumption (24 hours)
  - Power demand (city, country) (7 days)
  - Restaurant customers (7 days + 24 hours)
  - Ocean height (12.4 hours)
  - …



Cyclicality

- A **pattern recurrs at an unknown or variable interval.** $x_t \not\approx x_{t-S}$

- Examples:
  - Economic recession cycles
  - Climate phenomena (El Nino)
  - Machine operating cycles

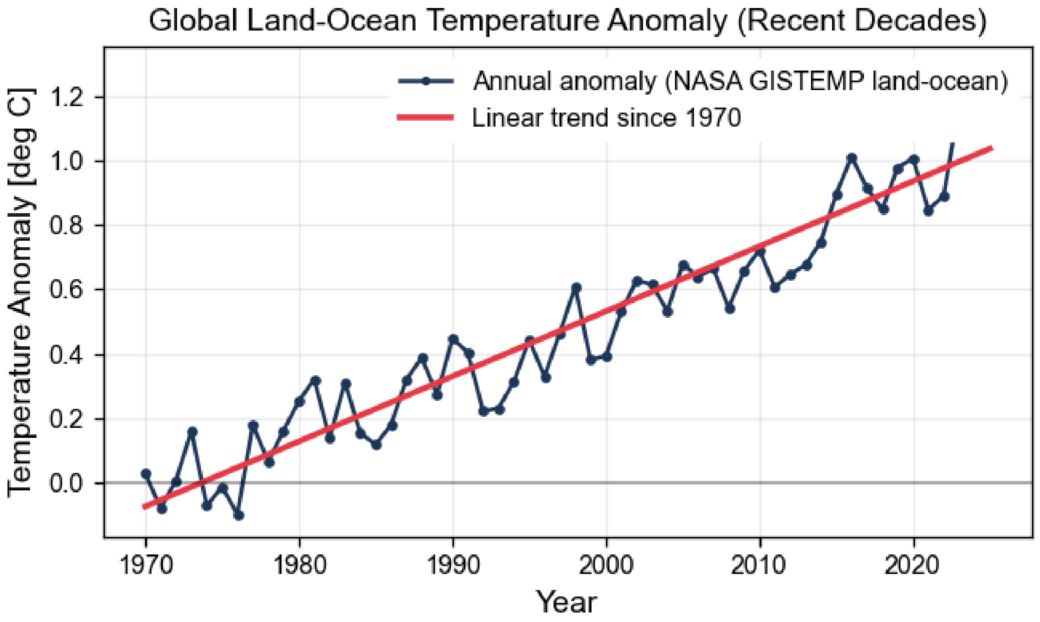


Trend

- **Trend**: gradual tendency of average series values to increase or decrease over time
  - Local trend: limited tendency over some time range
  - Global trend: tendency over complete measurement time range

- Different **forms of trends**:
  - Linear
  - Polynomial
  - Exponential
  - …

- **Examples**:
  - Global ocean temperatures (global warming)
  - Stock market indices
  - …



Irregular Temporal Dynamics

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- **Irregular dynamics**: temporal variations that have neither a persistent direction nor a reliably repeating pattern

- Examples:
  - Unpredictable fluctuations (turbulence)
  - Abrupt events or transients (financial returns)
  - Changing amplitudes and frequencies (weather variability)
  - Switching between operating regimes (machine faults)
  - Chaotic or stochastic behaviour (human activity)

- Irregular $\neq$ random!
  - Stochastic irregularity: generated partly by noise or random influences.
  - Chaotic irregularity: generated by a deterministic system with high sensitivity to its initial state.
  - Transient irregularity: caused by isolated events, faults, or regime changes.

Multiple scales

Slow dynamics

10^{-8} Hz

Seasonal & daily environmental loads
Temperature and humidity → $1.5 \cdot 10^{-8} - 1.1 \cdot 10^{-5}$ Hz

10^{-2} Hz

Global component wear
1 brake stop every 3 mins → 0.005 Hz

10^{0} Hz

Disk rotation
$1 - 180$ km/h → $0.14 - 26.3$ Hz (+ higher harmonics)

10^{1} Hz

10^{2} Hz

Stick-slip vibrations
Creep groan $100 - 500$ Hz

+

Brake system control
ESC, … → 250 Hz

10^{3} Hz

High frequency brake vibrations
Squeal $1 - 15$ kHz

10^{4} Hz

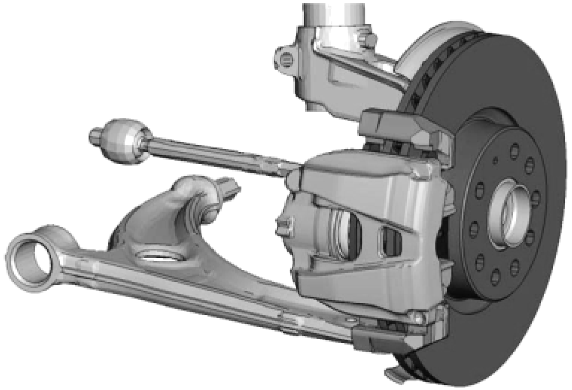
10^{5} Hz

10^{6} Hz

Asperity-level frictional processes
Raptures at atomic scale $0.1 - 100$ MHz

Fast dynamics

10^{9} Hz



- **Multi-scale dynamics**
- **Strong coupling between scales**
- **Non-stationary behavior**

Which time scales matter, how much of time history to take into account?

Deep Learning for TS

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Problem Statements

- **Time series classification**: recording of a bird's song → map to bird's species

- **Time series clustering** (unsupervised / self-supervised): anomaly detection

- **Time series regression**:
  - Sequence-to-sequence: virtual/soft sensor
  - Sequence-to-vector: remaining useful lifetime prediction, parameter identification

- **Time series forecasting**
  - Single-step
  - Multi-step

- **Approaches**:
  - Time-series only: models consume time series directly (or representations of it).
  - Featurization: tabular data of features is extracted from time series first

Feature Engineering

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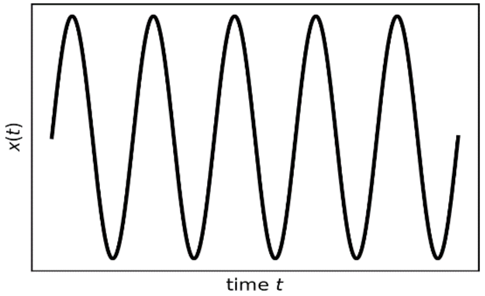
- **Machine learning**: structured data
- **Deep learning**: structured + unstructured data
- How to process unstructured engineering data (here: time series data) with ML?

$$[x(t_1) \quad \cdots \quad x(t_n)] \xrightarrow{\text{feature extraction}} [q_1 \quad q_2]$$

unstructured data
(time series)

structured data
(vector of features)



$$\longrightarrow [\max(x(t)) \quad \mathrm{mean}(x(t))]$$

- Requires domain expertise!

Example: Signal Classification

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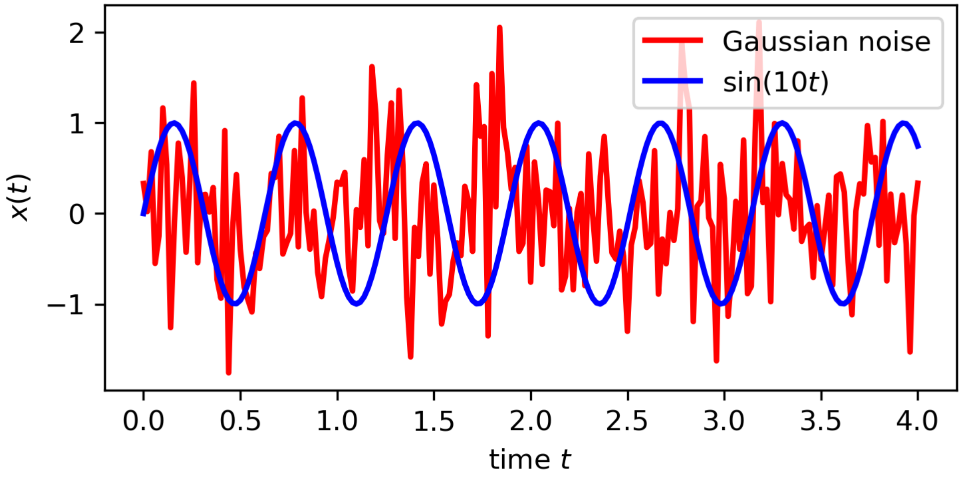
- Classify vibration data: a) **harmonic signal** and b) **random (Gaussian) process**
- Obvious to the human eye – how to teach the computer?
- **Without** feature engineering → deep learning required
- **With** feature engineering → simple decision tree



Statistical characterization

Not generic (amplitude / frequency variation?)

```
maximum of noise signal:        1.726001130529702
maximum of sine signal:         0.9999118601072672
mean of noise signal:           -1.7675192431843288e-17
mean of sine signal:            0.04318132362329466
median of noise signal:         0.017201635395270597
median of sine signal:          0.10071709699250761
standard dev. of noise signal:  0.7093145682637918
standard dev. of sine signal:   0.7093145682637918
```

Better: signal processing methods

FFT, Nonlinear Time Series Analysis

Time Series Classification

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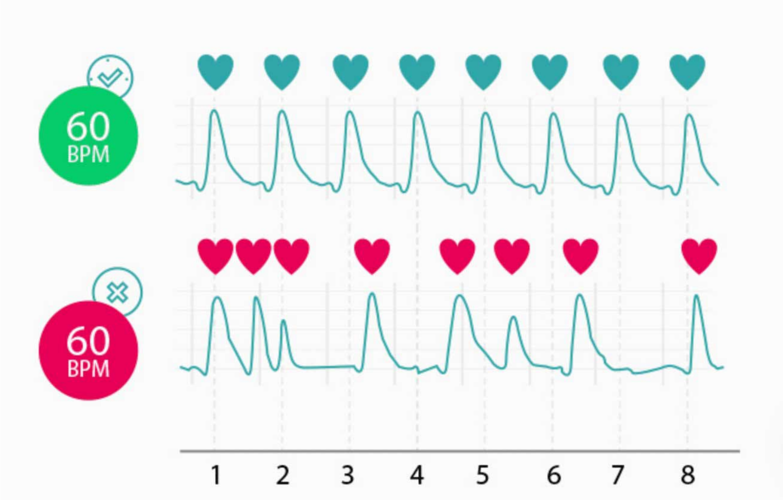
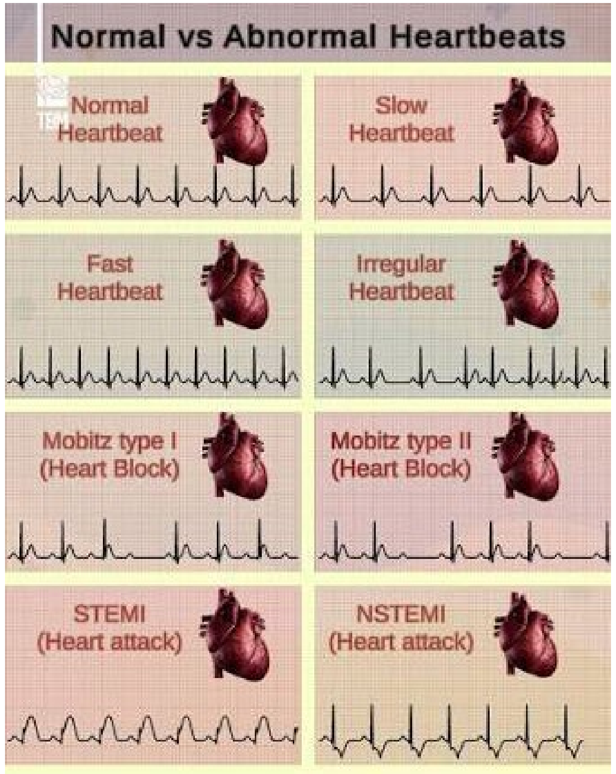
- Categorize time series



Cardiac anomolies (**unsupervised**)
https://www.fibricheck.com/heart-rate-versus-heart-rhythm/

Multi-class classification (**supervised**)
© thebrainmaze



Time Series Prediction

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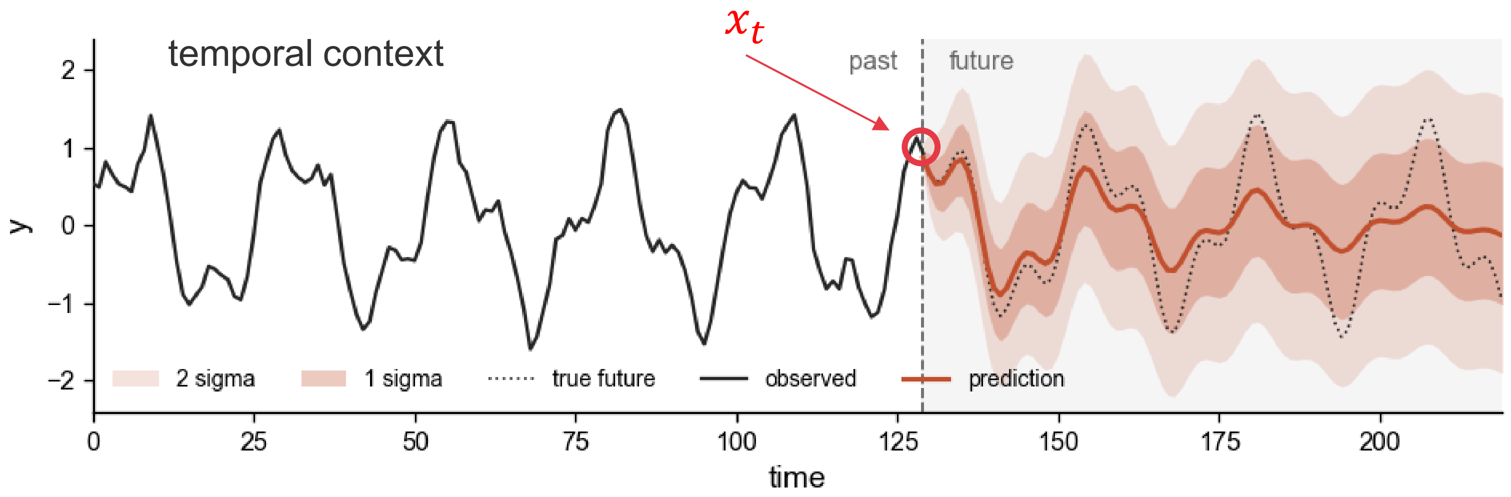
- Definition: estimate future values of one or more variables using observations available up to the present time.



Collapse to the mean?

- Setup: f_θ: $[x_{t-L+1}, x_2, \dots x_t] \mapsto [y_{t+1}, \dots y_{t+H}]$; general case: $\mathbb{R}^{L \times C_{\text{in}}} \mapsto \mathbb{R}^{H \times C_{\text{out}}}$
  - One-step forecasting: $H = 1$
  - Multi-step forecasting: $H > 1$
  - (+ auto-regression / self-feeding)

prediction horizon H [steps]
context length L [steps]

5min break

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Applications DL4TS

(from our research)

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Marine Learning and Decision Support

Ship-helicopter operation limits

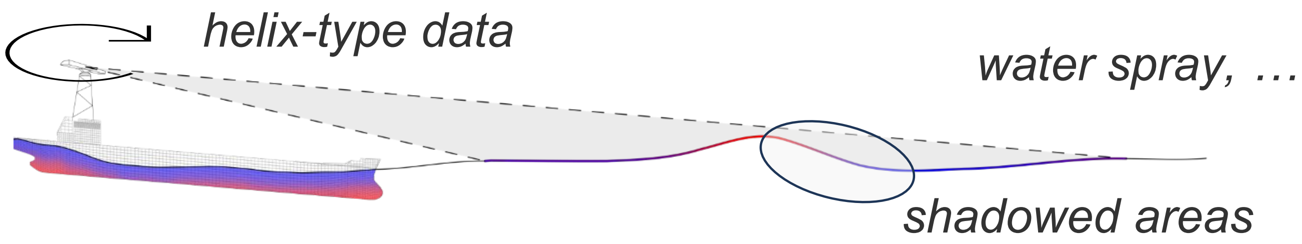
Ship-to-structure transfer

Marine construction

Short term (<5min) **wave forecasts** $\eta(x, y, t)$ required

- Radar: initial conditions $\eta(x, y, t_0)$
  noisy, corrupted, sparse

- Numerics: forecasting
  too slow



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Physics + Data + AI

Ocean Wave Forecasting

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- 1D waves $\mathbb{R}^{(8\times1024)} \mapsto \mathbb{R}^{1024}$

predicting >15 peak periods

auto-regressive set up

4000m spatial domain

- 2D waves

high-quality simulations

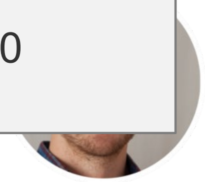
Models:
- U-Net (fully convolutional)
- 85k parameters (1D)
- 18k parameters (2D)

Data:
- 57 GB (1D)
- 3.4 TB (2D)

Training times
- ~30 hours (1D)*
- >1 week (2D)**

* Titan RTX3090
24GB VRAM

** Dual Nvidia A100
160GB VRAM



Learning Fields From Noisy Point Clouds

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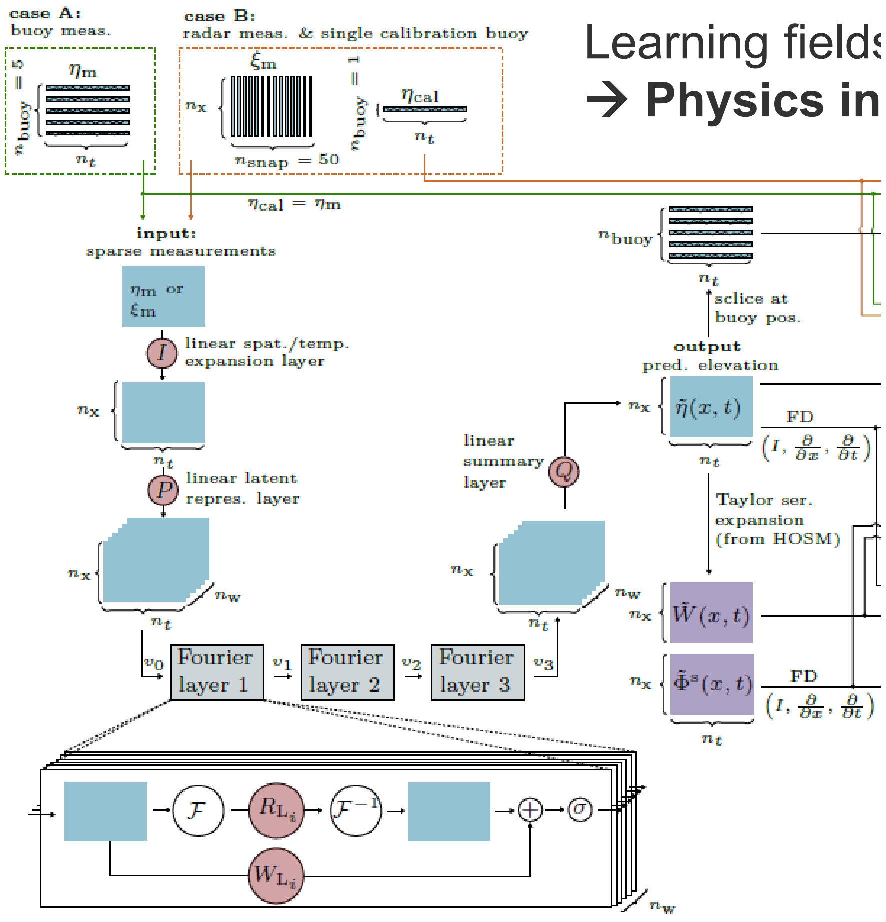
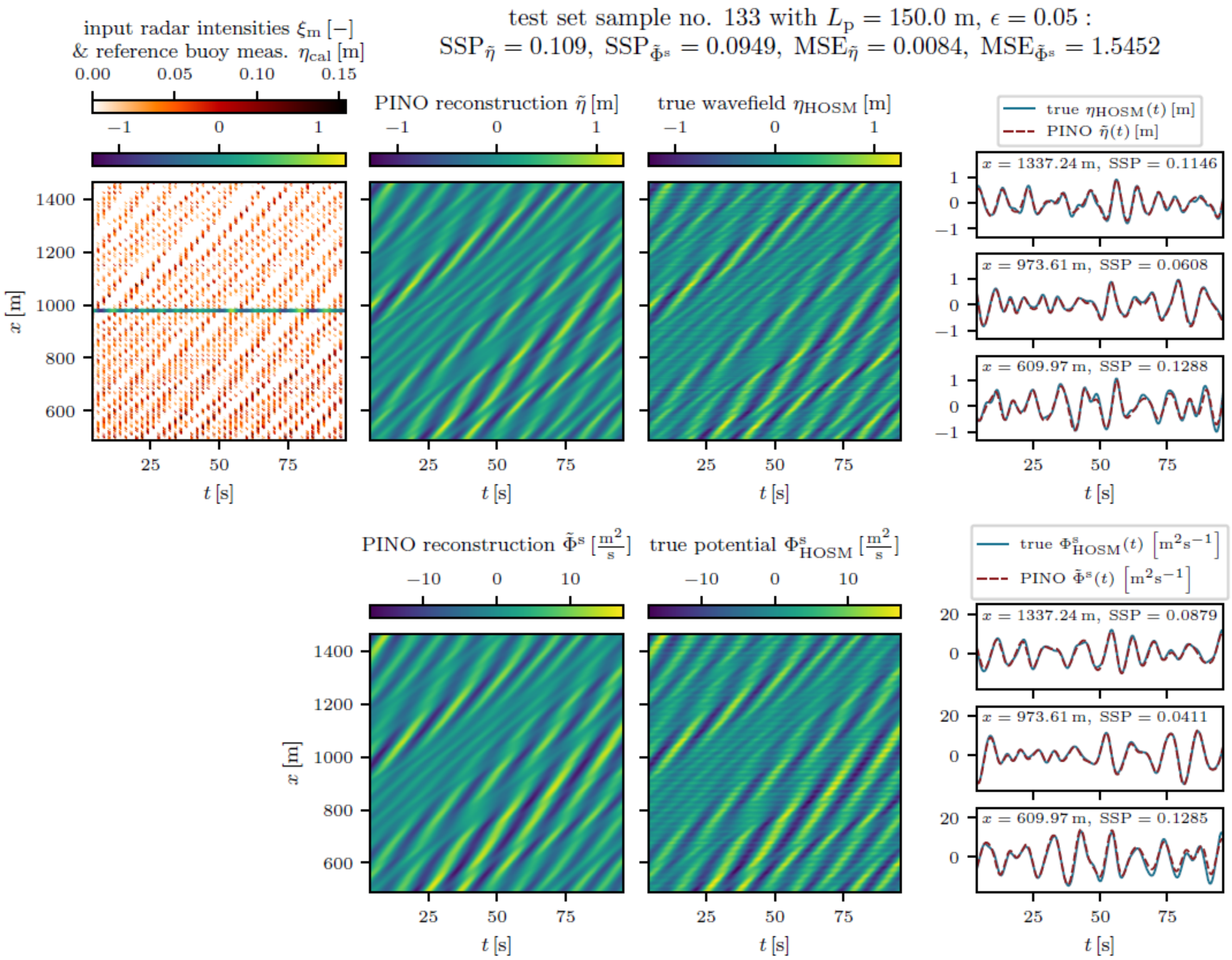
Learning fields
→ Physics in





Ehlers, Stender et al. (2023) *Ocean Engineering*
Ehlers, Stender et al. (2024) *Fluids*
Ehlers, Stender et al. (2025) *Physics of Fluids*

Physics + Data + AI

Differentiable Physics Simulations

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Core idea:

- Keeping a **physics-based mathematical structure**
- Adding **learnable terms / parts / modules**
- **End-to-end differentiation**

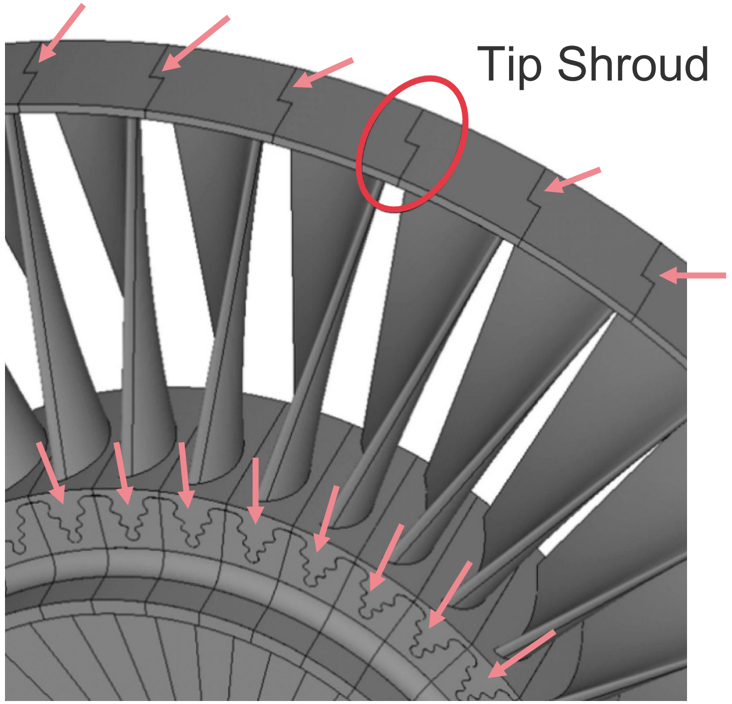

Image source: [https://doi.org/10.1016/j.ymssp.2022.110080]

Enables

- **Scalability** through vectorization
- **Explicit uncertainty** analysis (Monte-Carlo)
- **Learning unknown relationships** from **data***

*** scarce, largely unavailable**

(Vibration) Data is the New Oil

Advanced Time Series Analysis

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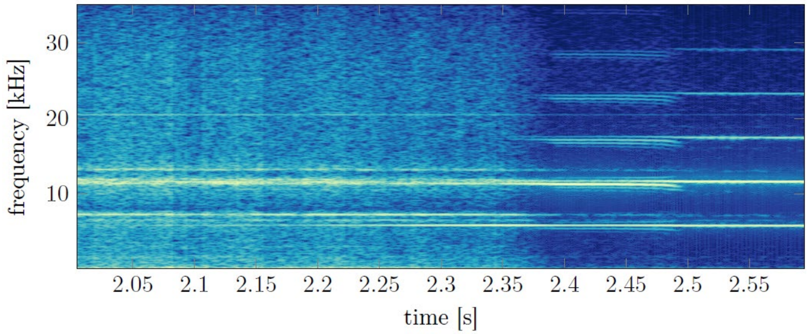
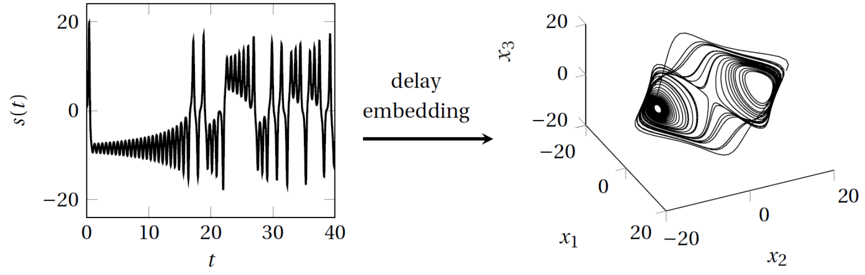
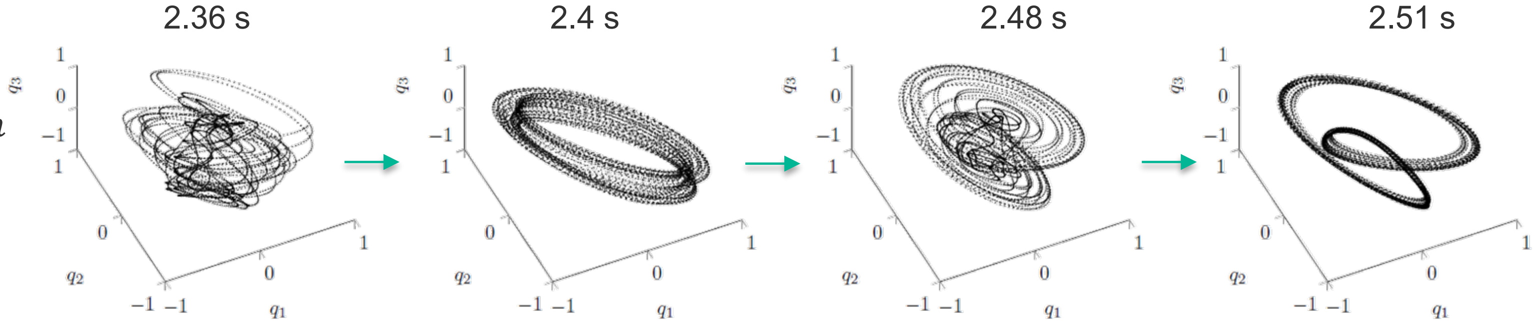
- Classical: FFT





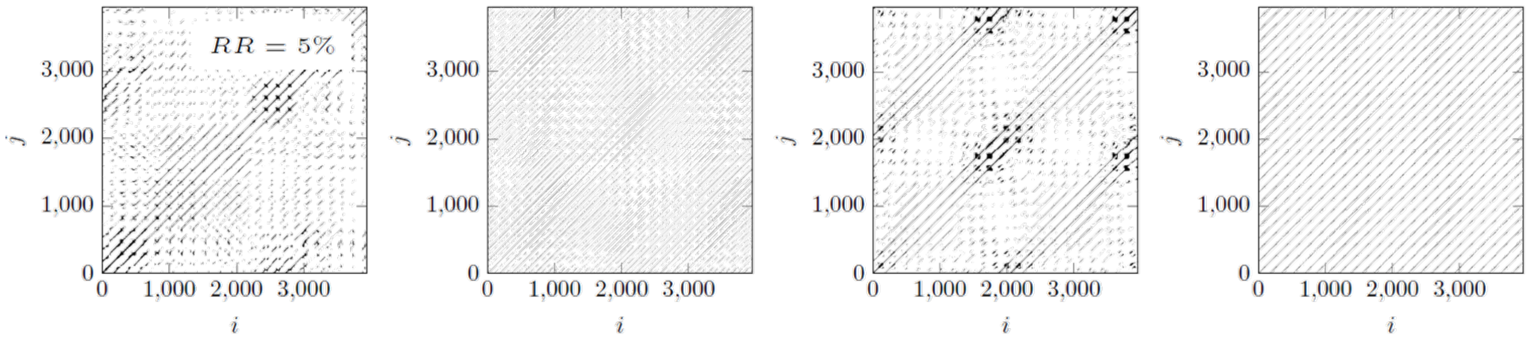
- Delay embedding (Takens' theorem)
  $x_n \equiv (s_n, \dots, s_{n+(m-1)k}) \in \mathbb{R}^m$



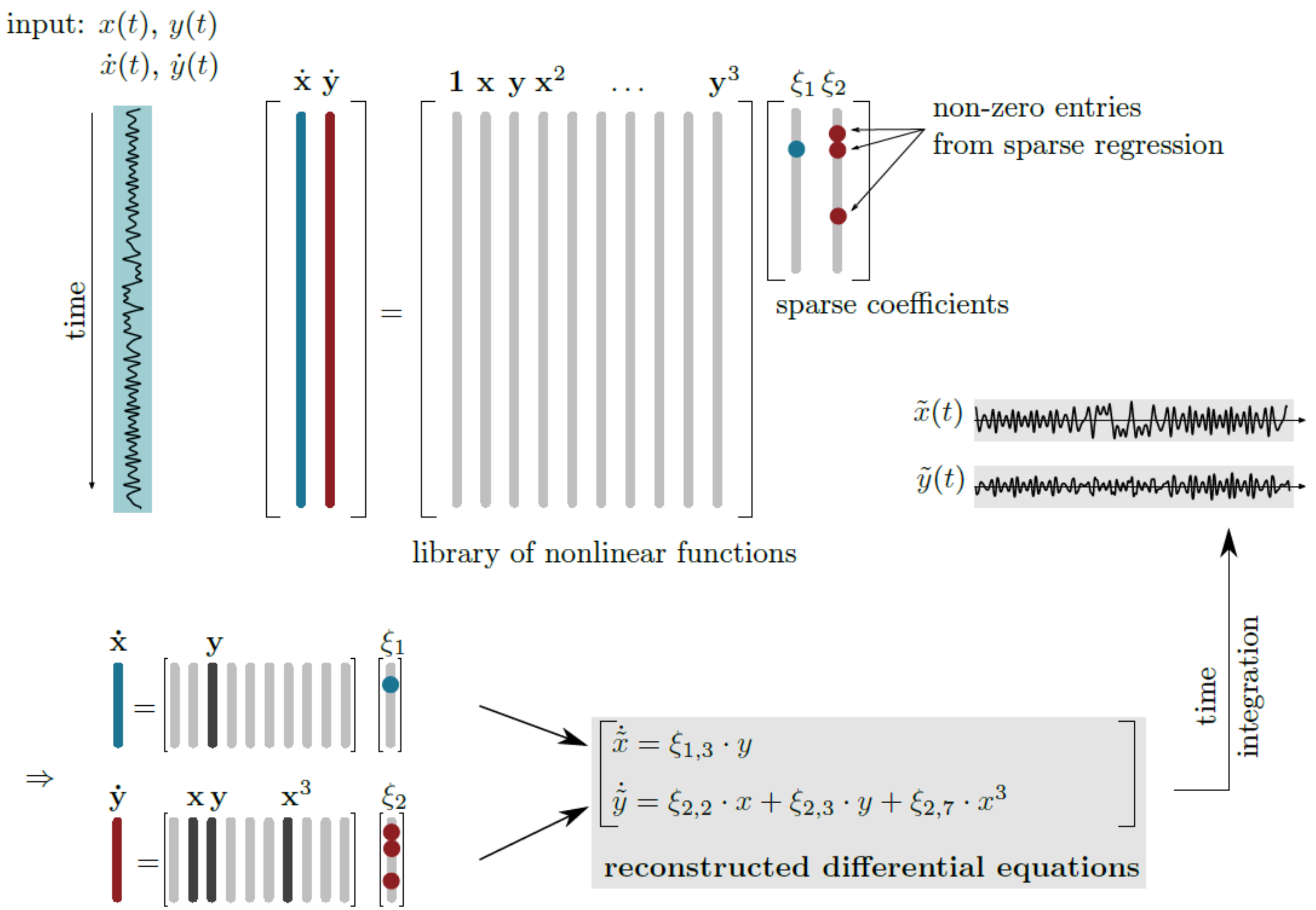
- Recurrence Plots

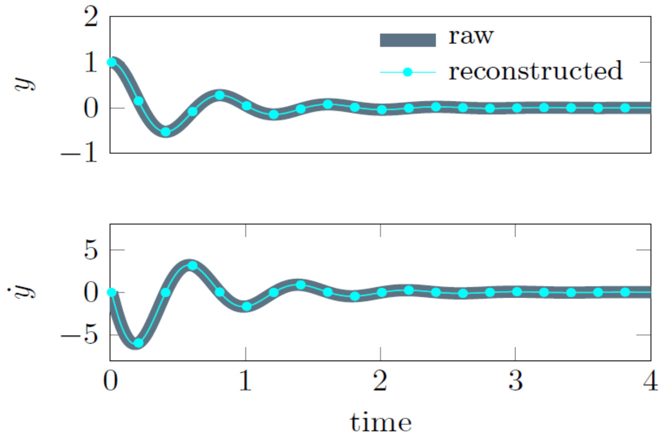
  $R_{i,j} = \Theta_\epsilon(|x_i - x_j|)$

  feature engineering



Discovering Equations from Data (SINDy)





Stender et al. (2019) *Vibration*

Illustrative Example

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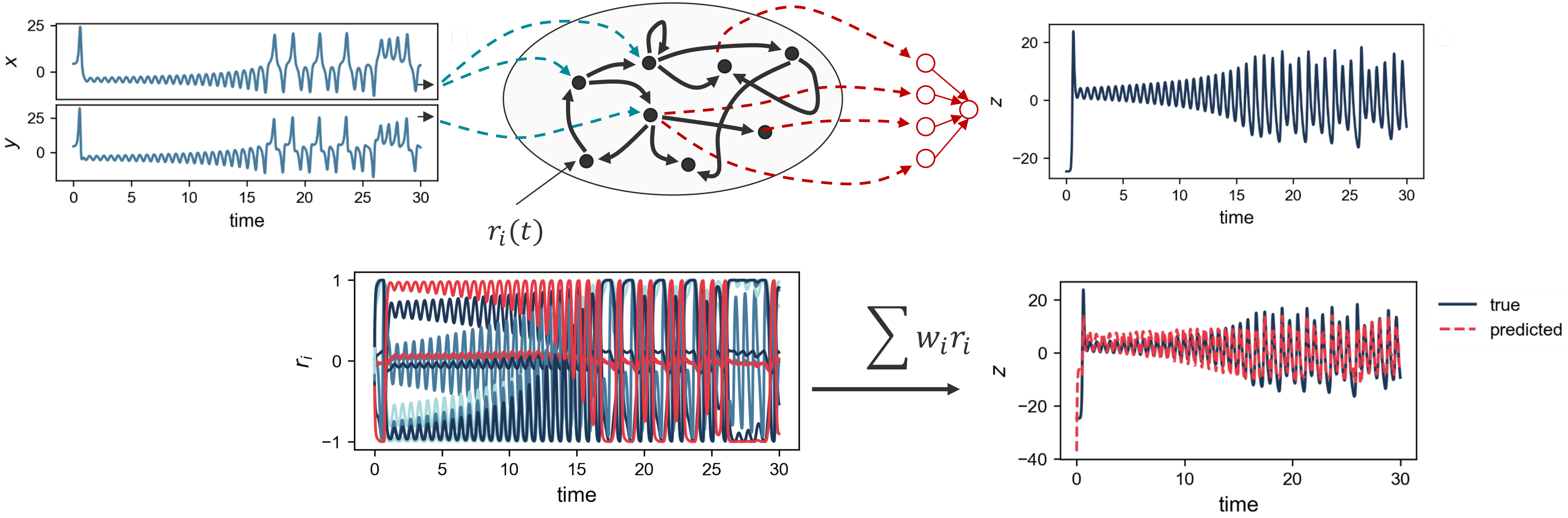
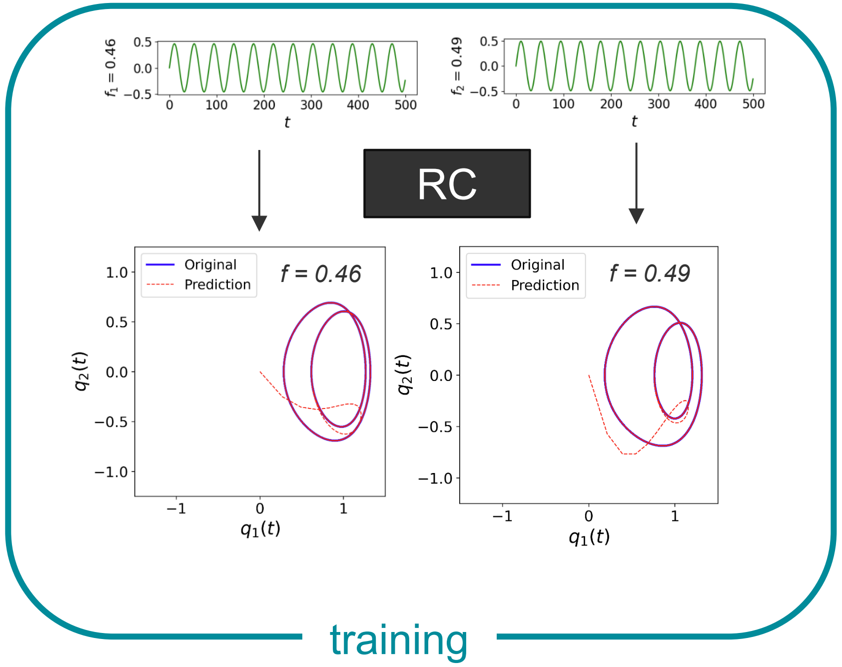
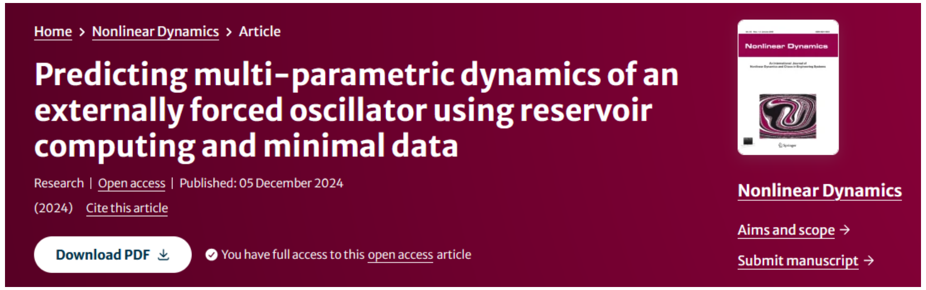
Reconstruction of an unobserved system state (Lorentz system)



Learning Bifurcating Dynamics

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Nonlinear Dynamics

Aims and scope →

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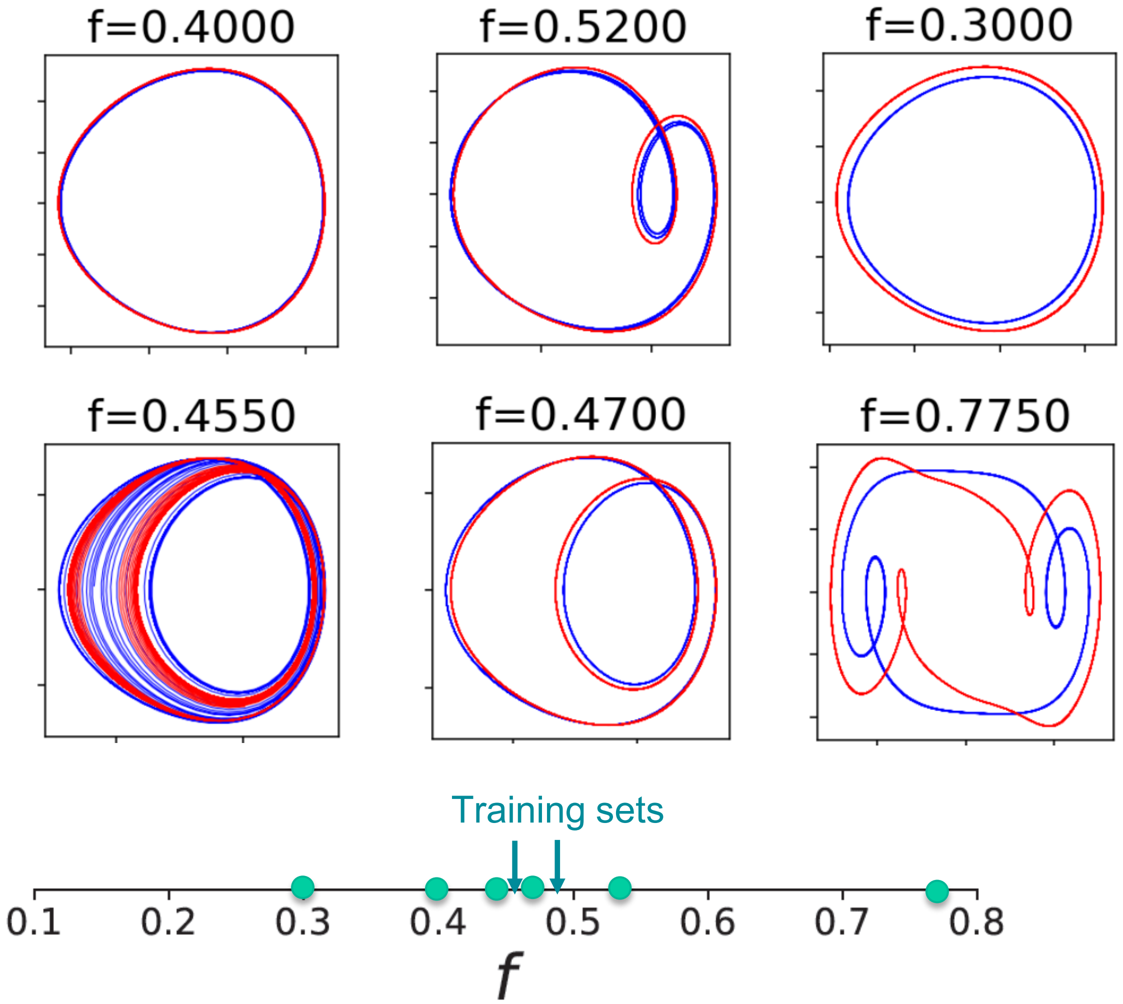

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Questions?

People with
- custom data sets
- special needs

Pls. come to front